ECA1403 EMBEDDED SYSTEMS

EXPERIMENT 2

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FORMAL DESIGN OF AN EMBEDDED CONTROLLER USING STATE MACHINE (STATECHART) MODELING – TRAFFIC LIGHT CONTROLLER

PROGRAME:

```
// ------------------------------------------------
// Formal Design of Embedded Controller Using FSM
// Traffic Light Controller
// Arduino UNO + Proteus
// ------------------------------------------------


enum TrafficState
{
  STATE_NS_GREEN,
  STATE_NS_YELLOW,
  STATE_EW_GREEN,
  STATE_EW_YELLOW
};

TrafficState currentState = STATE_NS_GREEN;

// North-South traffic signal
const int NS_RED_PIN    = 2;
const int NS_YELLOW_PIN = 3;
const int NS_GREEN_PIN  = 4;

// East-West traffic signal
const int EW_RED_PIN    = 5;
const int EW_YELLOW_PIN = 6;
const int EW_GREEN_PIN  = 7;
```

```
void setup()
{
  pinMode(NS_RED_PIN, OUTPUT);
  pinMode(NS_YELLOW_PIN, OUTPUT);
  pinMode(NS_GREEN_PIN, OUTPUT);

  pinMode(EW_RED_PIN, OUTPUT);
  pinMode(EW_YELLOW_PIN, OUTPUT);
  pinMode(EW_GREEN_PIN, OUTPUT);

  Serial.begin(9600);
}

void allOff()
{
  digitalWrite(NS_RED_PIN, LOW);
  digitalWrite(NS_YELLOW_PIN, LOW);
  digitalWrite(NS_GREEN_PIN, LOW);

  digitalWrite(EW_RED_PIN, LOW);
  digitalWrite(EW_YELLOW_PIN, LOW);
  digitalWrite(EW_GREEN_PIN, LOW);
}

void enterState(TrafficState state)
{
  allOff();

  switch (state)
  {
    case STATE_NS_GREEN:
```

```
  digitalWrite(NS_GREEN_PIN, HIGH);
  digitalWrite(EW_RED_PIN, HIGH);

  Serial.println("STATE S0: NS GREEN - EW RED");

  break;

case STATE_NS_YELLOW:

  digitalWrite(NS_YELLOW_PIN, HIGH);
  digitalWrite(EW_RED_PIN, HIGH);

  Serial.println("STATE S1: NS YELLOW - EW RED");

  break;

case STATE_EW_GREEN:

  digitalWrite(NS_RED_PIN, HIGH);
  digitalWrite(EW_GREEN_PIN, HIGH);

  Serial.println("STATE S2: NS RED - EW GREEN");

  break;

case STATE_EW_YELLOW:

  digitalWrite(NS_RED_PIN, HIGH);
  digitalWrite(EW_YELLOW_PIN, HIGH);

  Serial.println("STATE S3: NS RED - EW YELLOW");
```

```
      break;
  }
}

void loop()
{
  switch (currentState)
  {
    case STATE_NS_GREEN:

      enterState(STATE_NS_GREEN);
      delay(10000);
      currentState = STATE_NS_YELLOW;

      break;

    case STATE_NS_YELLOW:

      enterState(STATE_NS_YELLOW);
      delay(3000);
      currentState = STATE_EW_GREEN;

      break;

    case STATE_EW_GREEN:

      enterState(STATE_EW_GREEN);
      delay(10000);
      currentState = STATE_EW_YELLOW;

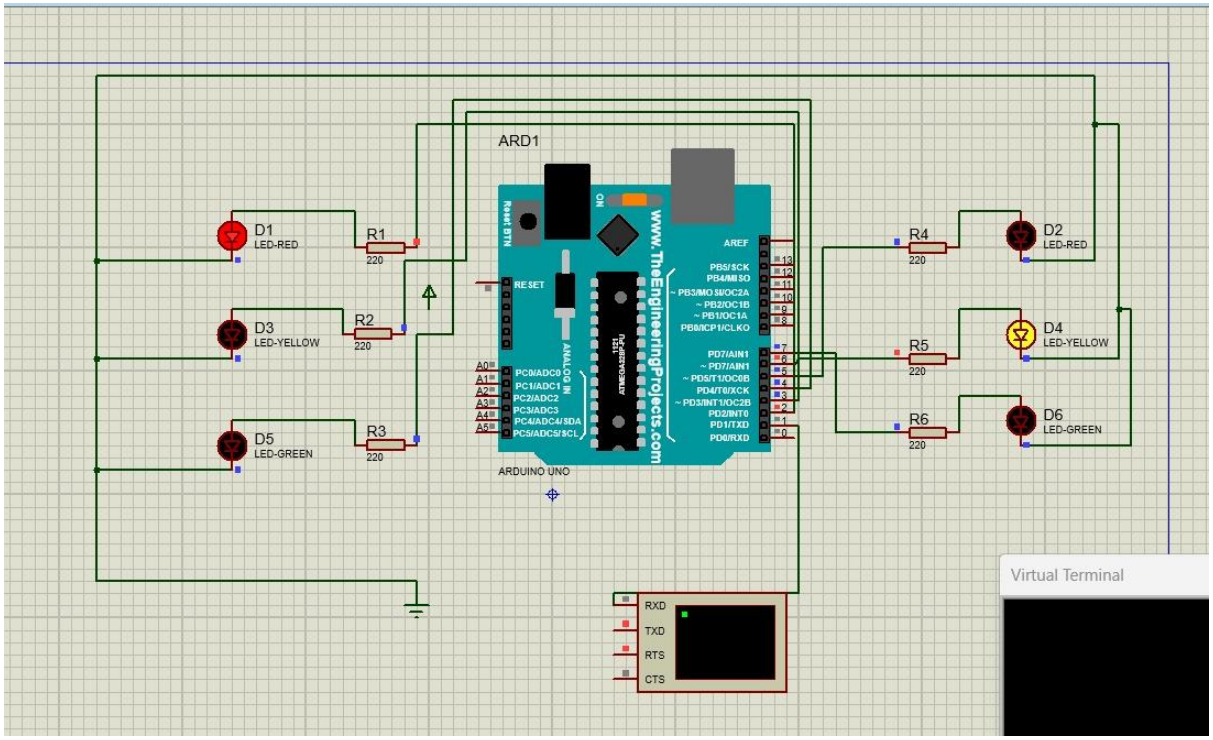
      break;
```

```
    case STATE_EW_YELLOW:

      enterState(STATE_EW_YELLOW);
      delay(3000);
      currentState = STATE_NS_GREEN;

      break;
  }
}
```

OUTPUT:



REFLECTIONS:

Through this experiment, I learned how to design and implement a traffic light controller using a Finite State Machine (FSM). I understood how states, transitions, and timing conditions control the traffic signals. I also learned to implement the controller using Arduino UNO and verify its operation through Proteus simulation and the Virtual Terminal.